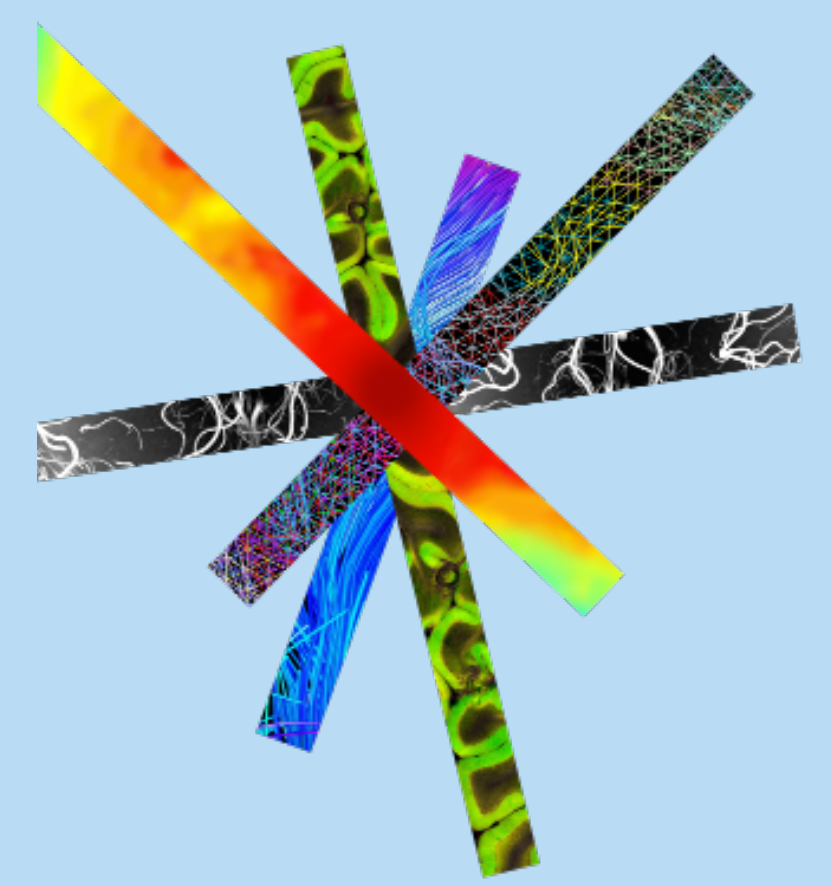




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INTRODUCTION

Major Depressive Disorder (MDD) is a widespread mental health condition involving persistent mood issues and cognitive challenges¹. While standard treatments are available, many patients do not fully recover, highlighting the need for alternative approaches. Transcranial photobiomodulation (t-PBM) is a novel therapy that uses non-ionizing light wavelengths (600–1100 nm) from sources like LEDs and lasers to stimulate brain function^{2,3}. This technique may enhance mitochondrial function, increase cellular energy production, and support neuroplasticity; however, the precise mechanisms underlying these effects remain unclear⁴.

MDD has been linked to volumetric changes in multiple brain regions critical to emotional regulation and cognition, such as the hippocampus, prefrontal cortex, amygdala, and basal ganglia^{5,6,7}. These changes are pronounced in patients with more severe or recurrent forms of MDD⁵. Evidence from longitudinal studies indicates that volumetric changes may be modifiable, with interventions promoting gray matter normalization in affected regions^{7,8}.

This study aims to examine whether t-PBM produces any volumetric changes in the brain.

METHODS

Preprocessing:

Structural T1 MRI images were preprocessed, segmented into WM, GM, and CSF images which were then normalized and modulated using CAT12's longitudinal segmentation pipeline optimized for capturing small structural changes.

Statistical Analysis:

To capture structural changes after t-PBM, we specified a flexible factorial model within CAT12 with the factors *subject*, *site*, and *time* with the *subject* factor representing the subject, *site* factor representing whether the site of the scan (MGH or NYU), and *time* factor representing the timing of the scan (first or last). We had a main effect of *subject* and an interaction effect of *site*time*. The model was estimated using Classical estimation (ReML) in SPM12. We used a contrast of -1 1 -1 1 with the design matrix. The resulting contrast image (thresholded at $\alpha = 0.05$, uncorrected) is shown within the results section.

Visualization:

The contrast was visualized using xjView toolbox (<https://www.alivelearn.net/xjview>) with cluster size of 50.

RESULTS

Analysis of T1-weighted MRI data revealed significant volumetric changes in the left cerebellum following t-PBM. Compared to baseline measurements, participants demonstrated a significant increase in left cerebellar volume ($p < 0.05$), after randomly receiving three different dosages. These findings suggest an effect of t-PBM on cerebellar structure.

No significant volumetric changes were observed in other brain regions, including cortical or subcortical structures. Exploratory whole-brain analyses confirmed the regional specificity of the observed changes, with significant effects localized exclusively to the left cerebellar hemisphere (Figure 1).

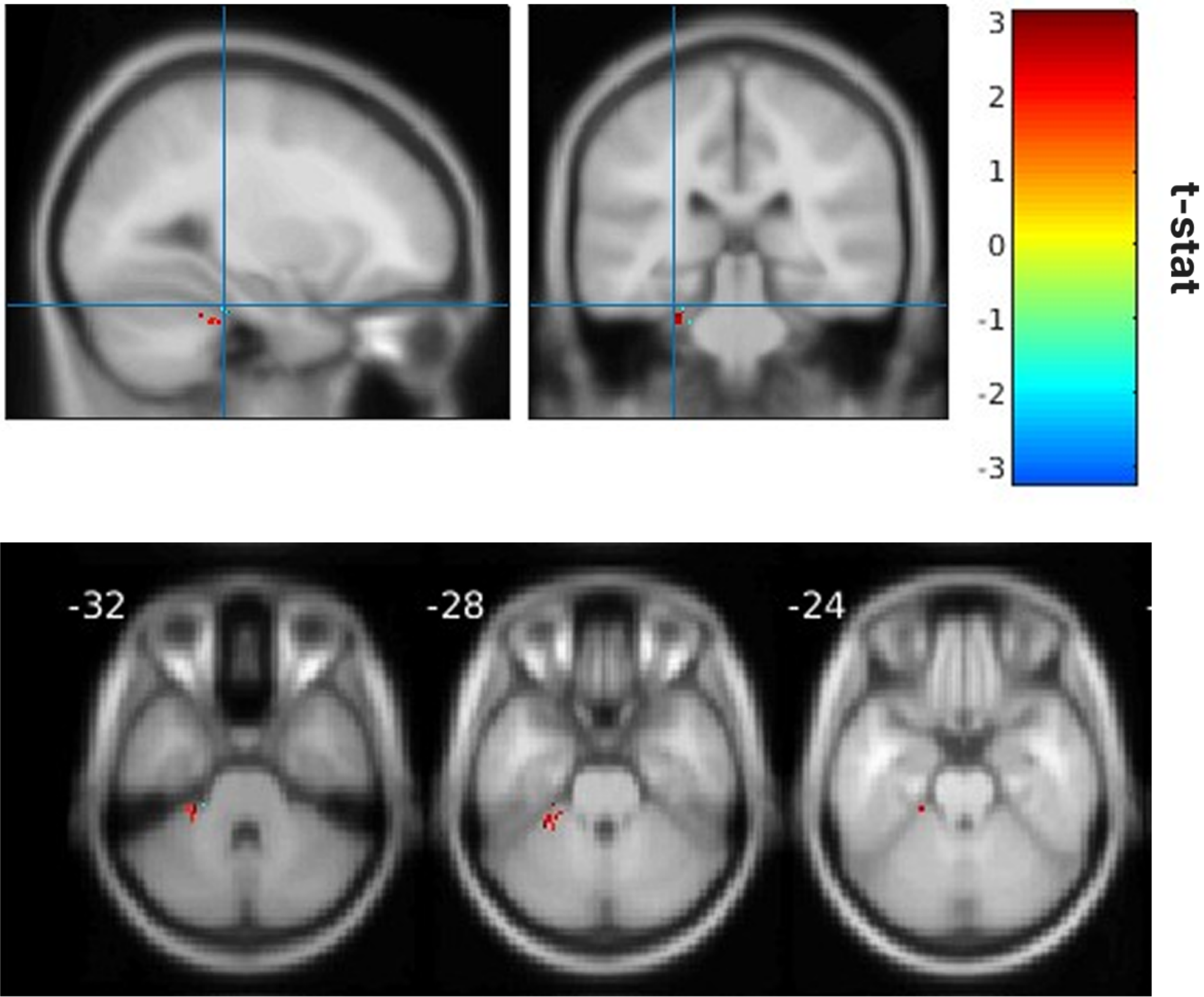


Figure 1: Sagittal, coronal, and transverse ($z = -32, -28, -24$ mm) views of volumetric changes. The t-contrast compares grey matter volume (GMV) between the first and last scan of the study.

#SUBJECT	AGE	SEX	RACE	ETHNICITY	SCREEN IDS	BL MADRS	BL SDQ
	Mean Age: 37.7 Median Age: 35 SD Age: 14.5	M: 8 F: 15	White: 16 Black: 2 Asian: 3 Mixed: 1 Other: 1	Hispanic: 2 Not Hispanic: 21	Mean IDS: 32.78 Median IDS: 31 SD IDS: 6.82	Mean MADRS: 27.47 Median MADRS: 29 SD MADRS: 6.014	Mean SDQ: 131.2 Median SDQ: 135 SD: 23.97

*Abbreviations: BL: Baseline, M: Male, F: Female, SD: Standard Deviation, IDS: Inventory for Depression Symptomatology, MADRS: Montgomery-Asberg Depression Rating Scale, SDQ: Symptoms of Depression Questionnaire

Table 1: Demographic information for the cohort included in this study

DISCUSSION

The cerebellum is associated with motor coordination and balance, it has also been recognized as a key player in emotional regulation and cognitive processing⁹. Recent research has highlighted its functional connectivity with limbic regions, including the prefrontal cortex and amygdala, which are central to mood regulation and highly implicated in the neuropathology of MDD¹⁰. Structural abnormalities in the cerebellum, including changes in the grey matter volume, have been reported in individuals with MDD¹¹. The observed volumetric changes in the left cerebellum in our study may reflect t-PBM-induced neuroplasticity, potentially mediated by its impact on mitochondrial function, cellular energy metabolism, and synaptic remodelling. These cellular effects are hypothesized to support the observed structural brain changes, possibly via mechanisms involving increased neurogenesis, gliogenesis, or dendritic spine formation. The cerebellum, with its high metabolic demand and rich synaptic architecture, may be particularly responsive to these effects.

The left-lateralized volumetric changes observed in the cerebellum raise intriguing questions about hemispheric asymmetry in t-PBM's neuromodulatory effects. Functional and structural asymmetries in the cerebellum have been reported in both healthy individuals and those with mood disorders, with the left cerebellum often showing greater involvement in cognitive and affective processes¹². It is possible that t-PBM preferentially affects the left cerebellum due to differences in connectivity, vascularization, or baseline metabolic activity, although further research is needed to elucidate this phenomenon

CONCLUSION

This study demonstrates that t-PBM induces volumetric changes in the left cerebellum in individuals with MDD. The findings suggest that t-PBM may promote neuroplasticity through mechanisms such as enhanced mitochondrial function and synaptic remodeling. The volumetric changes in the cerebellum highlight its role in emotional regulation and its potential as a target for neuromodulation. Future research is needed to explore the clinical significance of these changes and their correlation with symptoms improvement. These results provide preliminary support for t-PBM as a promising therapeutic intervention for MDD.

References

